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58 + U20=400V
59 + fn={fn1}
60 + N1=1000
61 + uR=0.04
62 + uX=0.08
63 + i0=0.05
64 + p0=0.005
65 + a={0.3} ; FS
66 + m={7} ; FS
67 + psi_0=0 ; Remanenzparameter zwischen 0,1,-1
68 + tpsi0=40ms
69 + tpsid=0.1ms
70 + ron=0.00001
71 + ctrl=1
72 .PARAM
73 + I1N={SN/U1N}
74 + Z1N={U1N/I1N}
75 + omegaN={{(2*pi)*fN}
76 + R1={Z1N*uR/2}
77 + L1s={Z1N*uX/2/omegaN}
78 + RFe={U1N*U1N/(p0*SN)}
79 + N2={N1*U20/U1N}
80 + R2={R1*N2/N1*N2/N1}
81 + Xh={{(U1N*U1N)/(SN*sqrt(i0*i0-p0*p0))}} ;FS
82 + iu={U1N/Xh} ;FS
83 + ImyN={sqrt(2)*iu} ;FS
84 + phiN={sqrt(2)*U1N/omegaN/N1}
85 + a={a} ; FS
86 + m={m} ; FS
87 * Windingscurrents
88 Gi1 1_1 1_2 VALUE={v(a,magnP1)/(N1*ctrl)}
89 Rpi1 1_1 1_2 1000MEG
90 Gi2 2_1 2_2 VALUE={v(b,a)/N2}
91 Rpi2 2_1 2_2 1000MEG
92 * Windings
93 X_W1 magnP1 a 1_1 1_2 winding_Yy
94 +PARAMS: R={R1} N={N1} Rphis={N1*N1/L1s}
95 +psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
96 X_W2 a b 2_1 2_2 winding_Yy
97 +PARAMS: R={R2} N={N2} Rphis={N1*N1/L1s}
98 +psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl=1
99 * Iron leg
100 X_SCHb magnP2 Leg_Yy PARAMS:
101 + N={N1} RFe={RFe} ImyN={ImyN} ctrl={ctrl}
102 + phiN={phiN} a={a} m={m}
103 .ENDS
104
105 .SUBCKT winding_Yy ma mb ea eb
106 +PARAMS: R=1 N=1 Rphis=1 psi_0=0 tpsi0=1000s tpsid=0.1ms ron=0.00001 ctrl=1
107 Gphi ma mb VALUE={v(phi)}
108 Rphis ma mb {Rphis*ctrl*ctrl}
109 *****
110 Gp 0 phi VALUE={v(ea,eb)-R*ctrl*v(mb,ma)/(N*ctrl))/(N*ctrl)}
111 Cp phi 0 1 IC=0
112 Rpphi 0 1000MEG
113 Vctrl ctrl 0 pwl 0s 0v {tpsi0+1p} 0v {tpsi0+tpsid+2p} 1V {tpsi0+2*tpsid+3p} 1v {tpsi0+3*tpsid+4p} 0v
114 S1 phi phix ctrl 0 MS1

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